

# SynDiff Synthetic Benchmark — Live Progress

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$S = I(X_1; Y | X_2) - I(X_1; Y)$  in nats;  $S > 0$  synergy-dominant,  $S < 0$  redundancy-dominant. SynDiff (ours) vs prior estimators.

## Dataset construction

### Phase 1 — Gaussian (closed-form ground truth)

Scalar base:  $X_1, X_2 \sim \mathcal{N}(0, 1)$  with  $\text{corr}(X_1, X_2) = \rho$ ; label  $Y = X_1 + X_2 + \sigma_y \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, 1)$ . Thus  $\text{Var}(Y) = 2 + 2\rho + \sigma_y^2$  and  $\text{Cov}(X_i, Y) = 1 + \rho$ . To lift to  $d$  dimensions the informative scalar is padded with  $d - 1$  i.i.d.  $\mathcal{N}(0, 1)$  coordinates and rotated by a fixed per-seed orthogonal matrix  $Q_i$  (one per variable). Rotation and independent padding preserve mutual information, so  $S$  is unchanged and the closed form carries over. Ground truth is computed from the  $3 \times 3$  covariance via  $I = \frac{1}{2} \log(\det \Sigma_A \det \Sigma_B / \det \Sigma_{AB})$  and the analogous conditional form. Cells sweep  $\rho \in \{-.9, \dots, .9\}$ ,  $\sigma_y \in \{.1, 1\}$ ,  $n \in \{1k, \dots, 100k\}$ ,  $d \in \{1, 8, 32, 128\}$ .

### Phase 2 — discrete U/R/S (mixture-density MC truth)

Latent sources  $s_1, s_2, s_3 \sim \text{Unif}\{0, \dots, Q - 1\}$  with  $Q = 8$ . Encodings  $X_i = W_i \text{onehot}(s_i) + \sigma_{\text{enc}} \mathcal{N}(0, I_{32})$  with orthonormal  $W_i \in \mathbb{R}^{32 \times 8}$  frozen per seed. Regimes ( $X_1$  always encodes  $s_1$ ): **UNIQUE**  $Y = s_1$ ; **REDUNDANT**  $Y = s_1$  and  $X_2$  also encodes  $s_1$ ; **SYNERGY1**  $Y = (s_1 + s_2) \bmod Q$  with  $X_2 \sim s_2$ ; **SYNERGY2**  $Y = (s_3 + (s_1 + s_2) \bmod Q) \bmod Q$  with  $X_2 = [\text{enc}(s_2) | \text{enc}(s_3)] \in \mathbb{R}^{64}$ ; **INDEPENDENT**  $Y = s_3$ . The  $\lambda$ -dial mixes the SYNERGY1 and REDUNDANT rules per sample (Bernoulli( $\lambda$ )). Ground truth is the exact  $S = I(X_1; X_2, Y) - I(X_1; X_2) - I(X_1; Y)$  obtained by Monte Carlo over the enumerable Gaussian-mixture components (matches the discrete limit as  $\sigma_{\text{enc}} \rightarrow 0$ :  $\pm \log Q$  for pure synergy/redundancy, 0 for unique/independent).

## Method: SynDiff

Target  $S = I(X_1; Y | X_2) - I(X_1; Y)$  in nats. By the symmetry of interaction information this equals  $I(X_1; X_2 | Y) - I(X_1; X_2)$ , which the four-mask form below estimates directly. **VP schedule**:  $\beta(t) = 0.1 + 19.9t$ ,  $B(t) = \int_0^t \beta = 0.1t + 9.95t^2$ , signal  $k(t) = e^{-B/2}$ , noise  $\sigma^2(t) = 1 - e^{-B}$ , weight  $w(t) = \beta(t)/(2\sigma^2(t))$ . A single noise-prediction network  $\hat{e}(x_{1,t}, t, \text{cond})$  (*MaskedCondMLP*: Fourier- $t$  features  $\oplus$  per-slot conditioning embeddings each with a learned NULL token;  $4 \times 512$  SiLU) is trained to denoise  $X_1$  while each conditioning slot ( $X_2, Y$ ) is independently dropped to its NULL token with probability 0.15. EMA weights (decay 0.999) are used at evaluation. **Four-mask estimator**. Per evaluation triple and per stratified time  $t$  ( $K=16$  strata on  $[10^{-3}, 1]$ ) with *one shared* noise draw  $\varepsilon$  and  $x_{1,t} = k(t)x_1 + \sigma(t)\varepsilon$ :

$$\begin{aligned} e_1 &= \hat{e}(x_{1,t}, t, x_2, y), & e_2 &= \hat{e}(x_{1,t}, t, \emptyset, y), \\ e_3 &= \hat{e}(x_{1,t}, t, x_2, \emptyset), & e_4 &= \hat{e}(x_{1,t}, t, \emptyset, \emptyset), \end{aligned}$$

$$\text{integrand}(t) = w(t)(\|e_1 - e_2\|^2 - \|e_3 - e_4\|^2).$$

Per-sample  $\hat{S} = (1 - 10^{-3}) \overline{\text{integrand}_t}$ ; the estimate is the mean over  $10^4$  fresh triples with a 1000-sample bootstrap CI (norms in fp32, no-grad, EMA weights). **Baselines**: *minde\_comp* (three

separately trained MINDE MI terms composed as above), *MINE* and *InfoNCE* (lower-bound MI, same composition), *batch\_pid* (Liang et al. CVX PID, k-means  $k=20$  then max-entropy solve), and *broja/plugin* on  $d=1$ .

## Sweep progress

| Phase | Method     | Done | Expected |
|-------|------------|------|----------|
| P1    | batch_pid  | 96   | 96       |
| P1    | broja      | 4    | 6        |
| P1    | infonce    | 96   | 96       |
| P1    | minde_comp | 96   | 96       |
| P1    | mine       | 96   | 96       |
| P1    | plugin     | 6    | 6        |
| P1    | syndiff    | 96   | 96       |
| P2    | batch_pid  | 90   | 90       |
| P2    | infonce    | 90   | 90       |
| P2    | minde_comp | 90   | 90       |
| P2    | mine       | 90   | 90       |
| P2    | syndiff    | 90   | 90       |

## Accuracy vs ground truth (completed cells, lower error better)

| Method         | #cells | mean $ \text{est} - \text{truth} $ | sign acc |
|----------------|--------|------------------------------------|----------|
| broja          | 2      | 0.0661                             | 1.00     |
| plugin         | 2      | 0.1301                             | 1.00     |
| minde_comp     | 62     | 0.1419                             | 0.81     |
| <b>syndiff</b> | 62     | <b>0.2277</b>                      | 0.89     |
| batch_pid      | 62     | 0.3660                             | 0.74     |
| mine           | 62     | 0.8132                             | 0.58     |
| infonce        | 62     | 3.9327                             | 0.37     |

## SynDiff vs prior — head to head (nats)

| Cell                         | truth  | <b>SynDiff</b> | batch_pid | minde_comp | mine   | infonce | plugin |
|------------------------------|--------|----------------|-----------|------------|--------|---------|--------|
| p1_dial_rho0_sy0p1_d8_n20k   | +1.963 | <b>+1.887</b>  | +0.147    | +1.922     | +0.631 | -5.604  | —      |
| p1_dial_rho0_sy1_d8_n20k     | +0.144 | <b>+0.136</b>  | +0.151    | +0.145     | -0.555 | -5.315  | —      |
| p1_dial_rho0p3_sy0p1_d8_n20k | +1.740 | <b>+1.677</b>  | +0.130    | +1.678     | +0.627 | -5.460  | —      |
| p1_dial_rho0p3_sy1_d8_n20k   | +0.007 | <b>+0.006</b>  | +0.128    | +0.010     | -0.603 | -5.429  | —      |
| p1_dial_rho0p5_sy0p1_d8_n20k | +1.477 | <b>+1.407</b>  | +0.111    | +1.434     | +0.525 | -5.436  | —      |
| p1_dial_rho0p5_sy1_d8_n20k   | -0.134 | <b>-0.134</b>  | +0.111    | -0.128     | -0.829 | -5.254  | —      |
| p1_dial_rho0p7_sy0p1_d8_n20k | +1.035 | <b>+0.971</b>  | +0.080    | +0.991     | +0.309 | -5.395  | —      |
| p1_dial_rho0p7_sy1_d8_n20k   | -0.329 | <b>-0.334</b>  | +0.088    | -0.321     | -0.651 | -5.190  | —      |
| p1_dial_rho0p9_sy0p1_d8_n20k | +0.024 | <b>-0.032</b>  | +0.048    | -0.025     | -0.329 | -5.321  | —      |
| p1_dial_rho0p9_sy1_d8_n20k   | -0.610 | <b>-0.601</b>  | +0.062    | -0.612     | -0.807 | -5.145  | —      |
| p1_dial_rho0p3_sy0p1_d8_n20k | +2.047 | <b>+1.976</b>  | +0.156    | +2.017     | +0.724 | -5.385  | —      |
| p1_dial_rho0p3_sy1_d8_n20k   | +0.209 | <b>+0.199</b>  | +0.156    | +0.210     | -0.562 | -5.363  | —      |

| Cell                          | truth  | SynDiff       | batch_pid | minde_comp | mine   | infonce | plugin |
|-------------------------------|--------|---------------|-----------|------------|--------|---------|--------|
| p1_dial_rhon0p5_sy0p1_d8_n20k | +2.023 | <b>+1.953</b> | +0.151    | +1.980     | +0.795 | -5.507  | -      |
| p1_dial_rhon0p5_sy1_d8_n20k   | +0.213 | <b>+0.197</b> | +0.152    | +0.214     | -0.504 | -5.342  | -      |
| p1_dial_rhon0p7_sy0p1_d8_n20k | +1.896 | <b>+1.826</b> | +0.136    | +1.853     | +0.689 | -5.488  | -      |
| p1_dial_rhon0p7_sy1_d8_n20k   | +0.177 | <b>+0.158</b> | +0.149    | +0.182     | -0.648 | -5.374  | -      |
| p1_dial_rhon0p9_sy0p1_d8_n20k | +1.473 | <b>+1.412</b> | +0.140    | +1.445     | +0.466 | -5.470  | -      |
| p1_dial_rhon0p9_sy1_d8_n20k   | +0.083 | <b>+0.054</b> | +0.143    | +0.089     | -0.580 | -5.200  | -      |
| p1_dim_rho0p7_sy1_d128_n20k   | -0.329 | <b>-0.050</b> | +0.140    | -0.295     | -1.852 | -6.133  | -      |
| p1_dim_rho0p7_sy1_d1_n20k     | -0.329 | <b>-0.329</b> | -0.250    | -0.322     | -0.321 | -0.271  | -0.191 |
| p1_dim_rho0p7_sy1_d32_n20k    | -0.329 | <b>-0.365</b> | +0.134    | -0.330     | -1.467 | -6.102  | -      |
| p1_dim_rho0p7_sy1_d128_n20k   | +0.177 | <b>+0.022</b> | +0.139    | +0.271     | -1.938 | -6.143  | -      |
| p1_dim_rho0p7_sy1_d1_n20k     | +0.177 | <b>+0.168</b> | +0.232    | +0.184     | +0.183 | +0.171  | +0.300 |
| p1_dim_rho0p7_sy1_d32_n20k    | +0.177 | <b>+0.129</b> | +0.146    | +0.177     | -1.085 | -6.105  | -      |
| p1_ns_rho0p7_sy1_d8_n100k     | -0.329 | <b>-0.332</b> | +0.089    | -0.321     | -0.704 | -5.204  | -      |
| p1_ns_rho0p7_sy1_d8_n1k       | -0.329 | <b>-0.331</b> | +0.089    | -0.325     | -0.630 | -5.217  | -      |
| p1_ns_rho0p7_sy1_d8_n20k      | -0.329 | <b>-0.333</b> | +0.089    | -0.322     | -0.652 | -5.261  | -      |
| p1_ns_rho0p7_sy1_d8_n5k       | -0.329 | <b>-0.331</b> | +0.089    | -0.321     | -0.651 | -5.174  | -      |
| p1_ns_rho0p7_sy1_d8_n100k     | +0.177 | <b>+0.157</b> | +0.149    | +0.180     | -0.648 | -5.374  | -      |
| p1_ns_rho0p7_sy1_d8_n1k       | +0.177 | <b>+0.160</b> | +0.149    | +0.183     | -0.497 | -5.338  | -      |
| p1_ns_rho0p7_sy1_d8_n20k      | +0.177 | <b>+0.154</b> | +0.149    | +0.180     | -0.602 | -5.375  | -      |
| p1_ns_rho0p7_sy1_d8_n5k       | +0.177 | <b>+0.154</b> | +0.149    | +0.185     | -0.453 | -5.449  | -      |
| p2_independent_senc0p1        | +0.000 | <b>+0.000</b> | +0.142    | -0.001     | -0.305 | -1.923  | -      |
| p2_independent_senc0p5        | +0.000 | <b>+0.000</b> | +0.141    | -0.000     | -0.763 | -2.083  | -      |
| p2_independent_senc1          | +0.000 | <b>-0.000</b> | +0.138    | -0.000     | -0.844 | -2.070  | -      |
| p2_lambda0_senc0p1            | -2.079 | <b>-6.167</b> | -2.054    | -2.590     | -2.063 | -2.191  | -      |
| p2_lambda0_senc0p5            | -0.730 | <b>-0.488</b> | -0.209    | -0.646     | -1.009 | -2.022  | -      |
| p2_lambda0_senc1              | -0.065 | <b>-0.000</b> | +0.144    | -0.049     | -0.975 | -2.085  | -      |
| p2_lambda0p25_senc0p1         | -0.302 | <b>-0.165</b> | -0.327    | +0.203     | -0.708 | -2.281  | -      |
| p2_lambda0p25_senc0p5         | -0.134 | <b>-0.137</b> | +0.027    | -0.223     | -0.760 | -2.062  | -      |
| p2_lambda0p25_senc1           | -0.016 | <b>+0.001</b> | +0.143    | -0.018     | -0.780 | -2.037  | -      |
| p2_lambda0p5_senc0p1          | +0.814 | <b>+1.008</b> | +0.794    | +1.347     | +0.138 | -2.142  | -      |
| p2_lambda0p5_senc0p5          | +0.272 | <b>+0.069</b> | +0.232    | +0.102     | -0.554 | -2.075  | -      |
| p2_lambda0p5_senc1            | +0.024 | <b>+0.004</b> | +0.144    | +0.004     | -0.953 | -2.079  | -      |
| p2_lambda0p75_senc0p1         | +1.583 | <b>+1.720</b> | +1.556    | +2.062     | +0.630 | -2.063  | -      |
| p2_lambda0p75_senc0p5         | +0.545 | <b>+0.333</b> | +0.376    | +0.391     | -0.817 | -2.090  | -      |
| p2_lambda0p75_senc1           | +0.048 | <b>+0.004</b> | +0.143    | +0.011     | -0.524 | -2.058  | -      |
| p2_lambda1_senc0p1            | +2.079 | <b>+2.284</b> | +2.066    | +2.826     | +0.830 | -2.160  | -      |
| p2_lambda1_senc0p5            | +0.733 | <b>+0.560</b> | +0.477    | +0.608     | -0.820 | -2.016  | -      |
| p2_lambda1_senc1              | +0.067 | <b>+0.000</b> | +0.145    | -0.000     | -0.796 | -2.101  | -      |
| p2_redundant_senc0p1          | -2.079 | <b>-5.757</b> | -2.052    | -3.164     | -2.058 | -2.194  | -      |
| p2_redundant_senc0p5          | -0.730 | <b>-0.425</b> | -0.207    | -0.635     | -0.924 | -2.035  | -      |
| p2_redundant_senc1            | -0.065 | <b>-0.000</b> | +0.134    | -0.041     | -0.768 | -2.071  | -      |
| p2_synergy1_senc0p1           | +2.079 | <b>+2.252</b> | +2.066    | +2.806     | +0.824 | -2.034  | -      |
| p2_synergy1_senc0p5           | +0.733 | <b>+0.546</b> | +0.482    | +0.608     | -0.918 | -2.094  | -      |
| p2_synergy1_senc1             | +0.067 | <b>-0.000</b> | +0.145    | -0.000     | -1.015 | -2.055  | -      |
| p2_synergy2_senc0p1           | +2.079 | <b>+0.000</b> | +0.718    | -0.001     | +0.623 | -1.968  | -      |
| p2_synergy2_senc0p5           | +0.424 | <b>-0.000</b> | +0.152    | -0.000     | -0.870 | -2.084  | -      |
| p2_synergy2_senc1             | +0.010 | <b>+0.000</b> | +0.142    | +0.000     | -0.748 | -2.064  | -      |
| p2_unique_senc0p1             | -0.000 | <b>-0.000</b> | +0.000    | +0.009     | -0.838 | -2.042  | -      |